# psymetadata: Social Identity Theory example

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```r
library(metaselection)
library(metadat)
library(psymetadata)
library(metafor)
```

```
## Loading required package: Matrix

## Loading required package: numDeriv

##
## Loading the 'metafor' package (version 4.8-0). For an
## introduction to the package please type: help(metafor)
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.3     v tidyr     1.3.2
## v purrr     1.2.1

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x tidyr::expand() masks Matrix::expand()
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## x tidyr::pack()   masks Matrix::pack()
## x tidyr::unpack() masks Matrix::unpack()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(here)
```

```
## here() starts at C:/Users/jamespustejovsky/Documents/metaselection
```

```r
devtools::load_all()
```

```
## i Loading metaselection
```

## Social Identity Theory Meta-analysis

```r
dat_ex <- steffens2020

dat_ex <- dat_ex |>
  rename(esid = es_id, studyid = study_id) |>
  mutate(
```

```r
    d = yi,
    Va = vi,
    Var_d = Va,
    sd_d = sqrt(Var_d),
    var_d = Var_d,
    p_onesided = 1 - pnorm(d / sd_d)
  )

length(unique(dat_ex$studyid)); length(unique(dat_ex$esid))
```

```
## [1] 128
```

```
## [1] 251
```

# Mean Only

## Unadjusted estimate: mean only

```r
ex_unadj_mean <- rma.uni(
  data = dat_ex,
  yi = d,
  sei = sd_d,
  method = "ML"
)
ex_unadj_mean
```

```
##
## Random-Effects Model (k = 251; tau^2 estimator: ML)
##
## tau^2 (estimated amount of total heterogeneity): 0.0858 (SE = 0.0083)
## tau (square root of estimated tau^2 value):      0.2929
## I^2 (total heterogeneity / total variability):   95.96%
## H^2 (total variability / sampling variability):  24.77
##
## Test for Heterogeneity:
## Q(df = 250) = 6351.3935, p-val < .0001
##
## Model Results:
##
## estimate      se     zval    pval   ci.lb   ci.ub
##   0.4151  0.0193  21.5529  <.0001  0.3773  0.4528  ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

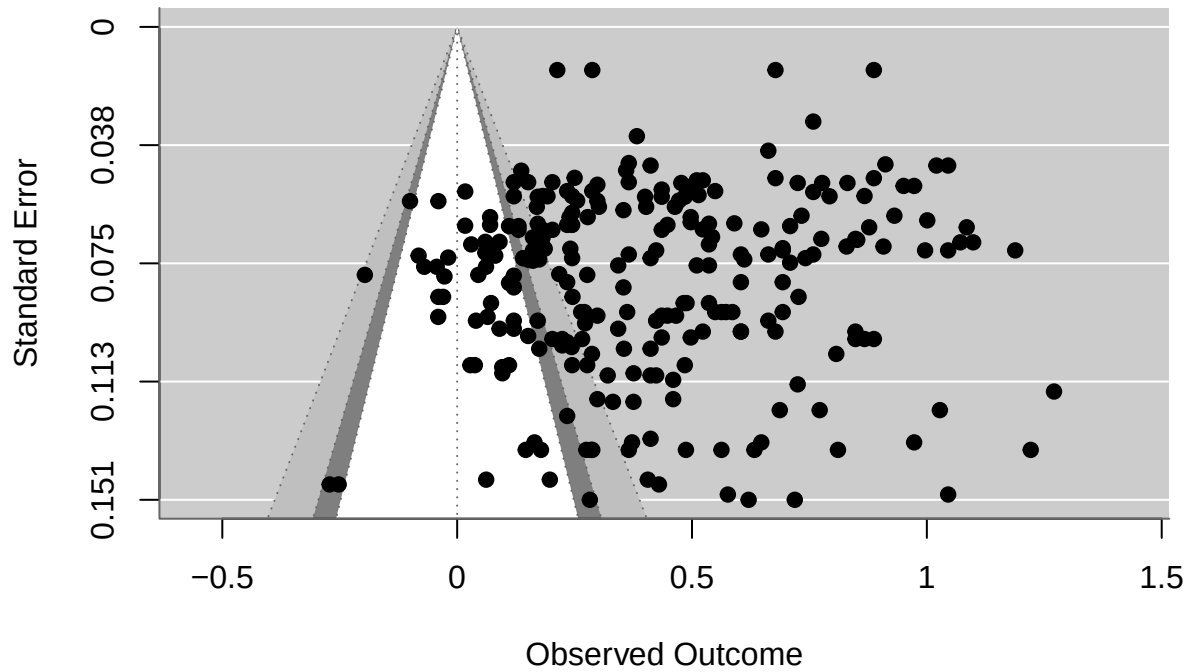
```r
ex_unadj_mean_cil <- ex_unadj_mean$tau2 - 1.96*ex_unadj_mean$se.tau2
ex_unadj_mean_ciu <- ex_unadj_mean$tau2 + 1.96*ex_unadj_mean$se.tau2
ex_unadj_mean_cil; ex_unadj_mean_ciu
```

```
## [1] 0.06954973
```

```
## [1] 0.1020752
```

### Forest plot

```
funnel(ex_unadj_mean, refline = 0, level = c(90, 95, 99))
```




### CHE estimate: mean only

```
V_mat <- vcalc(
    vi = vi,
    cluster = studyid,
    obs = esid,
    data = dat_ex,
    rho = 0.5,
    sparse = TRUE
  )

ex_che <- rma.mv(yi = yi,
                 V = V_mat,
                 random = ~ 1 | studyid / esid,
                 data = dat_ex,
                 sparse = TRUE,
                 digits = 3) |>
    robust(cluster = studyid, clubSandwich = TRUE)

ex_che
```

```
##
```

```
## Multivariate Meta-Analysis Model (k = 251; method: REML)
##
## Variance Components:
##
##            estim   sqrt  nlvls  fixed        factor
## sigma^2.1  0.054  0.233    128     no       studyid
## sigma^2.2  0.034  0.185    251     no  studyid/esid
##
## Test for Heterogeneity:
## Q(df = 250) = 7669.807, p-val < .001
##
## Number of estimates:   251
## Number of clusters:    128
## Estimates per cluster: 1-6 (mean: 1.96, median: 2)
##
## Model Results:
##
## estimate     se¹    tval¹      df¹   pval¹  ci.lb¹  ci.ub¹
##    0.404  0.025   16.005   124.32   <.001   0.354   0.454   ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## 1) results based on cluster-robust inference (var-cov estimator: CR2,
##    approx t-test and confidence interval, df: Satterthwaite approx)
```

### Adjusted estimate in metaselection: Beta model mean only

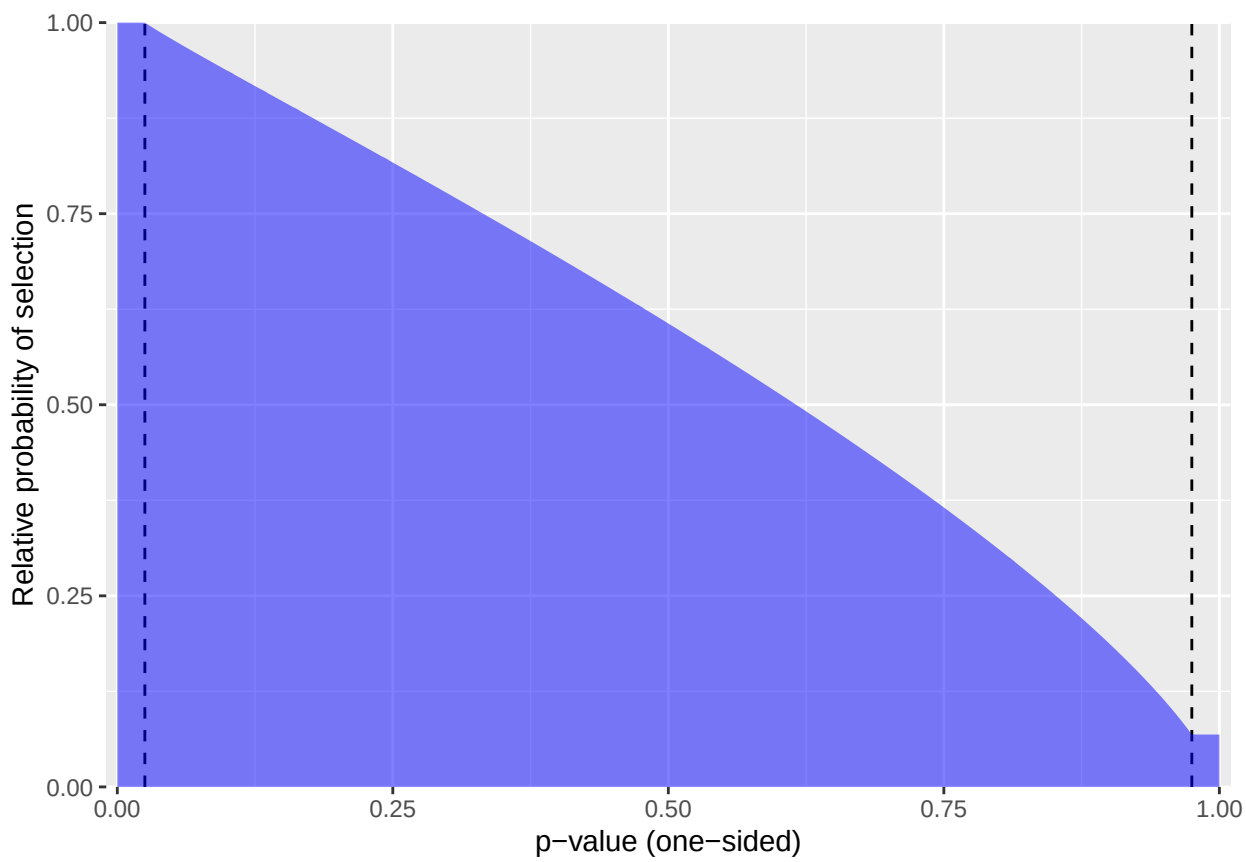
```r
ex_adj_ms_beta_mean <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  pi = p_onesided,
  cluster = studyid,
  selection_type = "beta",
  steps = c(.025,.975),
  make_sandwich = TRUE
)

summary(ex_adj_ms_beta_mean)
```

```
## Beta Density Model
##
## Call:
## selection_model(data = dat_ex, yi = d, sei = sd_d, pi = p_onesided,
##     cluster = studyid, selection_type = "beta", steps = c(0.025,
##         0.975), make_sandwich = TRUE)
##
## Number of clusters = 128; Number of effects = 251
##
## Steps: 0.025, 0.975
## Estimator: composite marginal likelihood
## Variance estimator: robust
##
```

```
## Log composite likelihood of selection model: -49.0215
##
## Mean effect estimates:
##                                    Large Sample
##  Coef. Estimate Std. Error  p-value Lower  Upper
##   beta    0.279     0.0806 0.000534 0.121  0.437
##
## Heterogeneity estimates:
##                                    Large Sample
##  Coef. Estimate Std. Error p-value  Lower  Upper
##   tau2    0.142     0.0255     --- 0.0997  0.202
##
## Selection process estimates:
##                                      Large Sample
##    Coef. Estimate Std. Error  p-value Lower  Upper
##  lambda1    0.995      0.139 0.971388 0.756   1.31
##  lambda2    1.727      0.249 0.000152 1.302   2.29
```

### Selection plot: Beta model mean only

```r
selection_plot(ex_adj_ms_beta_mean)
```



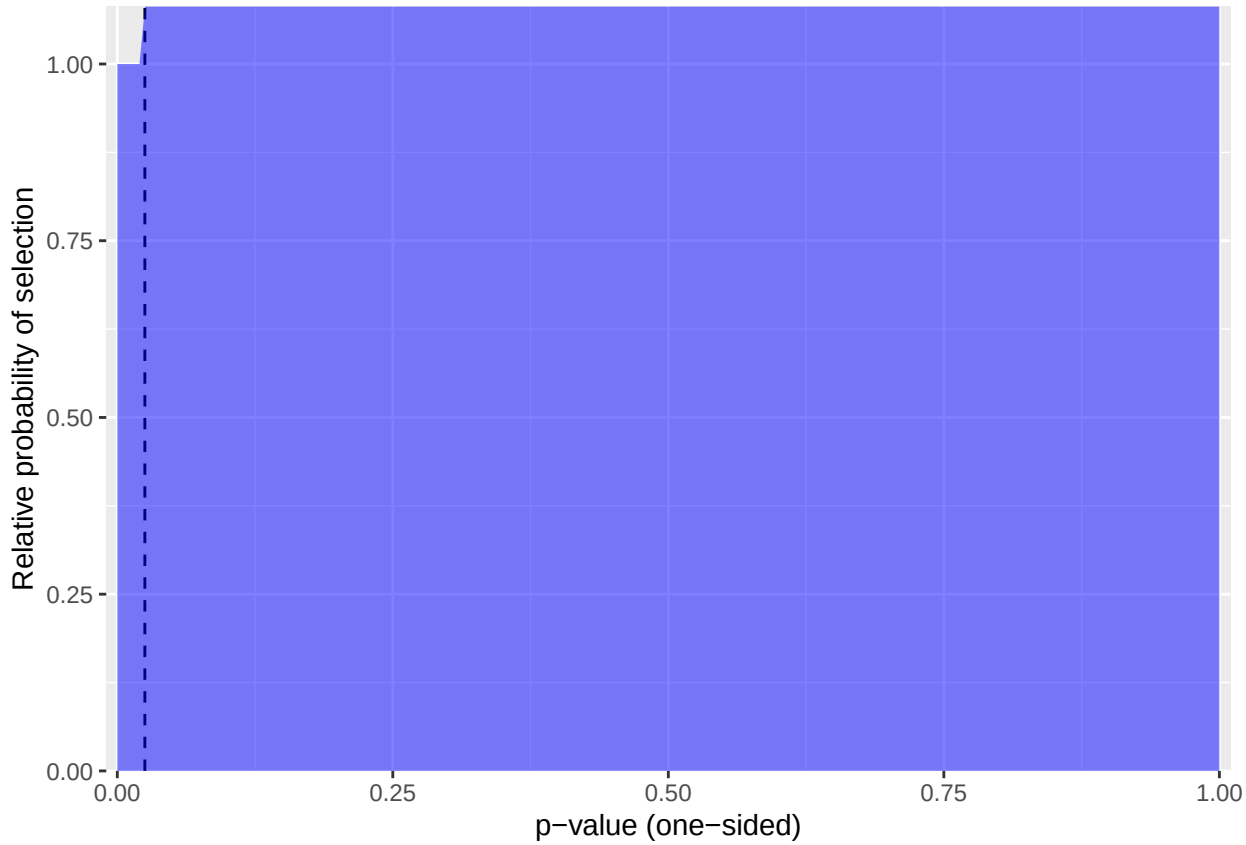

### Adjusted estimate in metaselection: Step 3PSM model mean only

```r
ex_adj_ms_step3_mean <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = 0.025
)

summary(ex_adj_ms_step3_mean)
```

```
## Step Function Model
##
## Call:
## selection_model(data = dat_ex, yi = d, sei = sd_d, cluster = studyid,
##     selection_type = "step", steps = 0.025)
##
## Number of clusters = 128; Number of effects = 251
##
## Steps: 0.025
## Estimator: composite marginal likelihood
## Variance estimator: robust
##
## Log composite likelihood of selection model: -58.12295
## Inverse selection weighted partial log likelihood: 170.7777
##
## Mean effect estimates:
##                                     Large Sample
##  Coef. Estimate Std. Error  p-value Lower  Upper
##   beta    0.421     0.0345 3.06e-34 0.354  0.489
##
## Heterogeneity estimates:
##                                     Large Sample
##  Coef. Estimate Std. Error p-value  Lower  Upper
##   tau2    0.084    0.00982     --- 0.0668  0.106
##
## Selection process estimates:
##  Step: 0 < p <= 0.025; Studies: 106; Effects: 198
##                                      Large Sample
##    Coef. Estimate Std. Error p-value Lower  Upper
##  lambda0        1        ---     ---   ---    ---
##
##  Step: 0.025 < p <= 1; Studies: 38; Effects: 53
##                                      Large Sample
##    Coef. Estimate Std. Error p-value Lower  Upper
##  lambda1     1.08      0.324   0.794 0.601   1.94
```

### Selection plot: Step 3 PSM model mean only

```r
selection_plot(ex_adj_ms_step3_mean)
```




### Adjusted estimate in metaselection: Step 4PSM model mean only

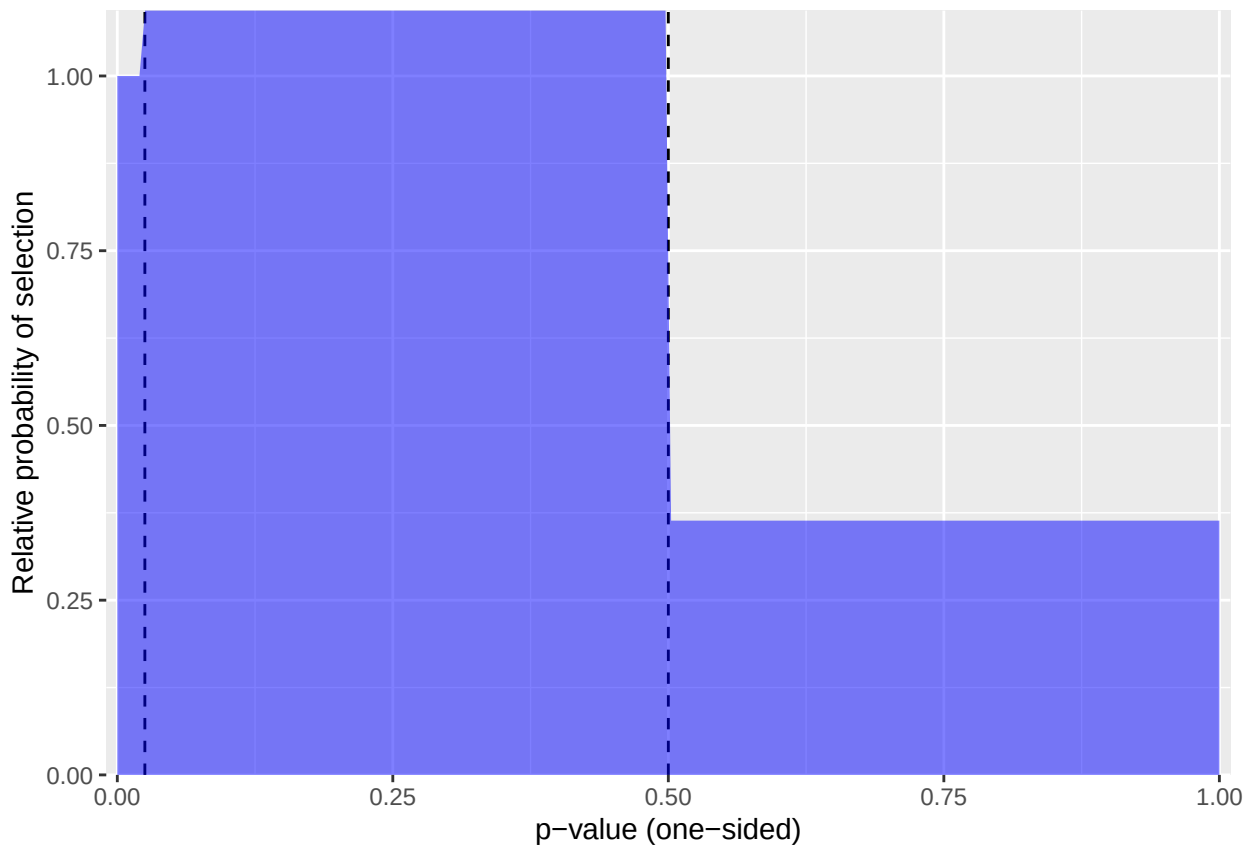
```
ex_adj_ms_step4_mean <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = c(0.025, .5)
)

summary(ex_adj_ms_step4_mean)
```

```
## Step Function Model
##
## Call:
## selection_model(data = dat_ex, yi = d, sei = sd_d, cluster = studyid,
##     selection_type = "step", steps = c(0.025, 0.5))
##
## Number of clusters = 128; Number of effects = 251
##
## Steps: 0.025, 0.5
## Estimator: composite marginal likelihood
## Variance estimator: robust
##
## Log composite likelihood of selection model: -52.93508
```

```
## Inverse selection weighted partial log likelihood: 168.0726
##
## Mean effect estimates:
##                                     Large Sample
##  Coef. Estimate Std. Error  p-value Lower  Upper
##   beta    0.367     0.0462 2.07e-15 0.276  0.457
##
## Heterogeneity estimates:
##                                     Large Sample
##  Coef. Estimate Std. Error p-value  Lower  Upper
##   tau2    0.107     0.0143     --- 0.0826  0.139
##
## Selection process estimates:
##  Step: 0 < p <= 0.025; Studies: 106; Effects: 198
##                                      Large Sample
##    Coef. Estimate Std. Error p-value Lower  Upper
##  lambda0        1        ---     ---   ---    ---
##
##  Step: 0.025 < p <= 0.5; Studies: 32; Effects: 40
##                                      Large Sample
##    Coef. Estimate Std. Error p-value Lower  Upper
##  lambda1     1.09      0.272   0.719 0.672   1.78
##
##  Step: 0.5 < p <= 1; Studies: 9; Effects: 13
##                                      Large Sample
##    Coef. Estimate Std. Error p-value Lower  Upper
##  lambda2    0.364      0.186  0.0479 0.134  0.991
```

### Selection plot: Step 4PSM model mean only

```r
selection_plot(ex_adj_ms_step4_mean)
```




# Moderator

## Unadjusted estimate: moderator

```
ex_unadj_mod <- rma.uni(
  data = dat_ex,
  yi = d,
  sei = sd_d,
  method = "ML",
  mods = ~ design
)
ex_unadj_mod
```

```
##
## Mixed-Effects Model (k = 251; tau^2 estimator: ML)
##
## tau^2 (estimated amount of residual heterogeneity):     0.0743 (SE = 0.0073)
## tau (square root of estimated tau^2 value):             0.2726
## I^2 (residual heterogeneity / unaccounted variability): 95.35%
## H^2 (unaccounted variability / sampling variability):   21.52
## R^2 (amount of heterogeneity accounted for):            13.39%
##
## Test for Residual Heterogeneity:
## QE(df = 249) = 5627.9474, p-val < .0001
##
```

```
## Test of Moderators (coefficient 2):
## QM(df = 1) = 34.9749, p-val < .0001
##
## Model Results:
##
##          estimate      se    zval    pval   ci.lb   ci.ub
## intrcpt    0.2544  0.0326  7.7991  <.0001  0.1905  0.3183  ***
## design     0.2315  0.0391  5.9140  <.0001  0.1547  0.3082  ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
confint(ex_unadj_mod)
```

```
##
##        estimate   ci.lb   ci.ub
## tau^2    0.0743  0.0623  0.0916
## tau      0.2726  0.2496  0.3026
## I^2(%)  95.3539 94.5058 96.1967
## H^2     21.5235 18.2011 26.2932
```

### Adjusted estimate in metaselection: Beta model moderator

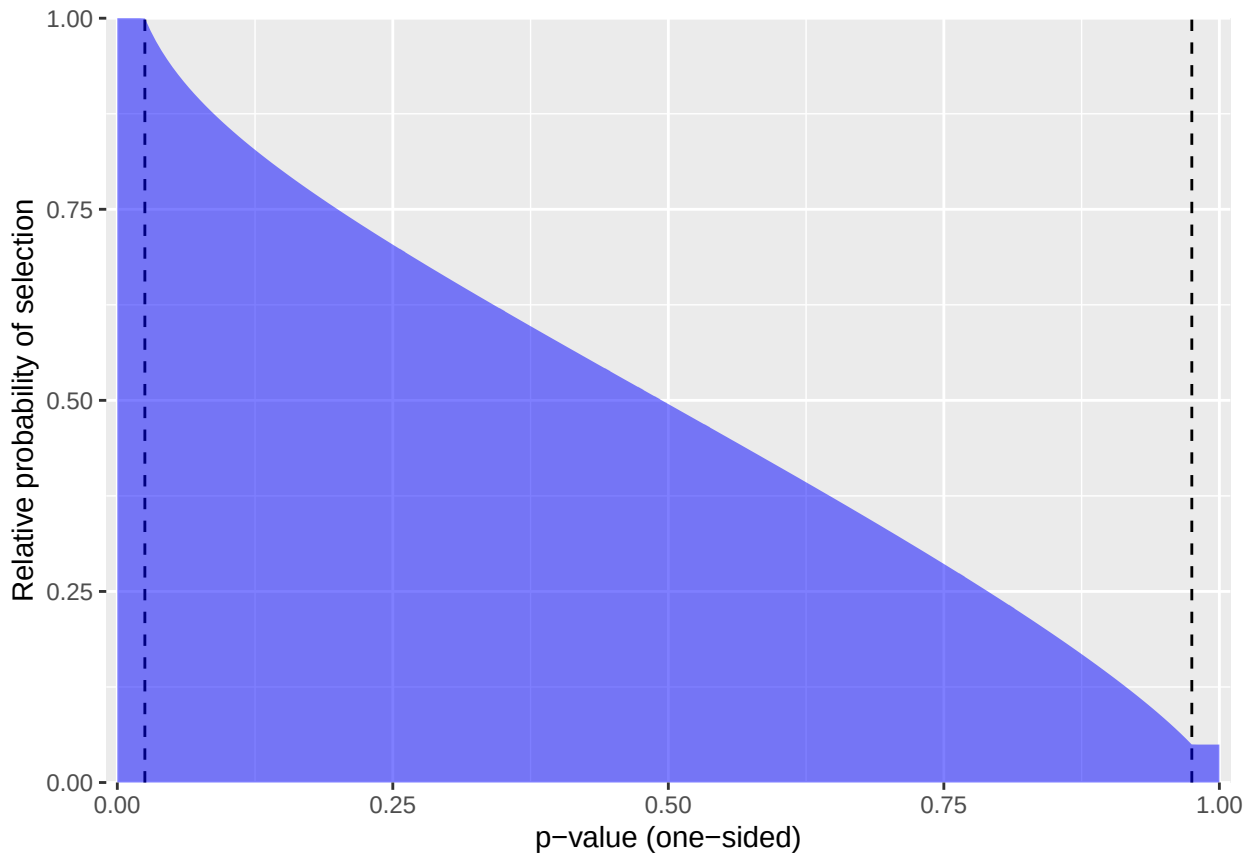
```r
ex_adj_ms_beta_mod <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  pi = p_onesided,
  cluster = studyid,
  selection_type = "beta",
  steps = c(.025,.975),
  make_sandwich = TRUE,
  mean_mods = ~ design
)
summary(ex_adj_ms_beta_mod)
```

```
## Beta Density Model
##
## Call:
## selection_model(data = dat_ex, yi = d, sei = sd_d, pi = p_onesided,
##     cluster = studyid, selection_type = "beta", steps = c(0.025,
##         0.975), mean_mods = ~design, make_sandwich = TRUE)
##
## Number of clusters = 128; Number of effects = 251
##
## Steps: 0.025, 0.975
## Estimator: composite marginal likelihood
## Variance estimator: robust
##
## Log composite likelihood of selection model: -31.48084
##
## Mean effect estimates:
##                                                Large Sample
##             Coef. Estimate Std. Error  p-value  Lower  Upper
##  beta_(Intercept)  0.00374      0.119 0.974993 -0.230  0.237
```

```
##       beta_design  0.39877      0.106 0.000173  0.191  0.607
##
## Heterogeneity estimates:
##                                     Large Sample
##  Coef. Estimate Std. Error p-value  Lower  Upper
##   tau2    0.119     0.0191     --- 0.0868  0.163
##
## Selection process estimates:
##                                       Large Sample
##    Coef. Estimate Std. Error  p-value Lower  Upper
##  lambda1    0.933      0.135 0.632080 0.703   1.24
##  lambda2    1.753      0.270 0.000271 1.296   2.37
```

### Selection plot: Beta model moderator

```
selection_plot(ex_adj_ms_beta_mod)
```




### Adjusted estimate in metaselection: Step 3PSM model moderator

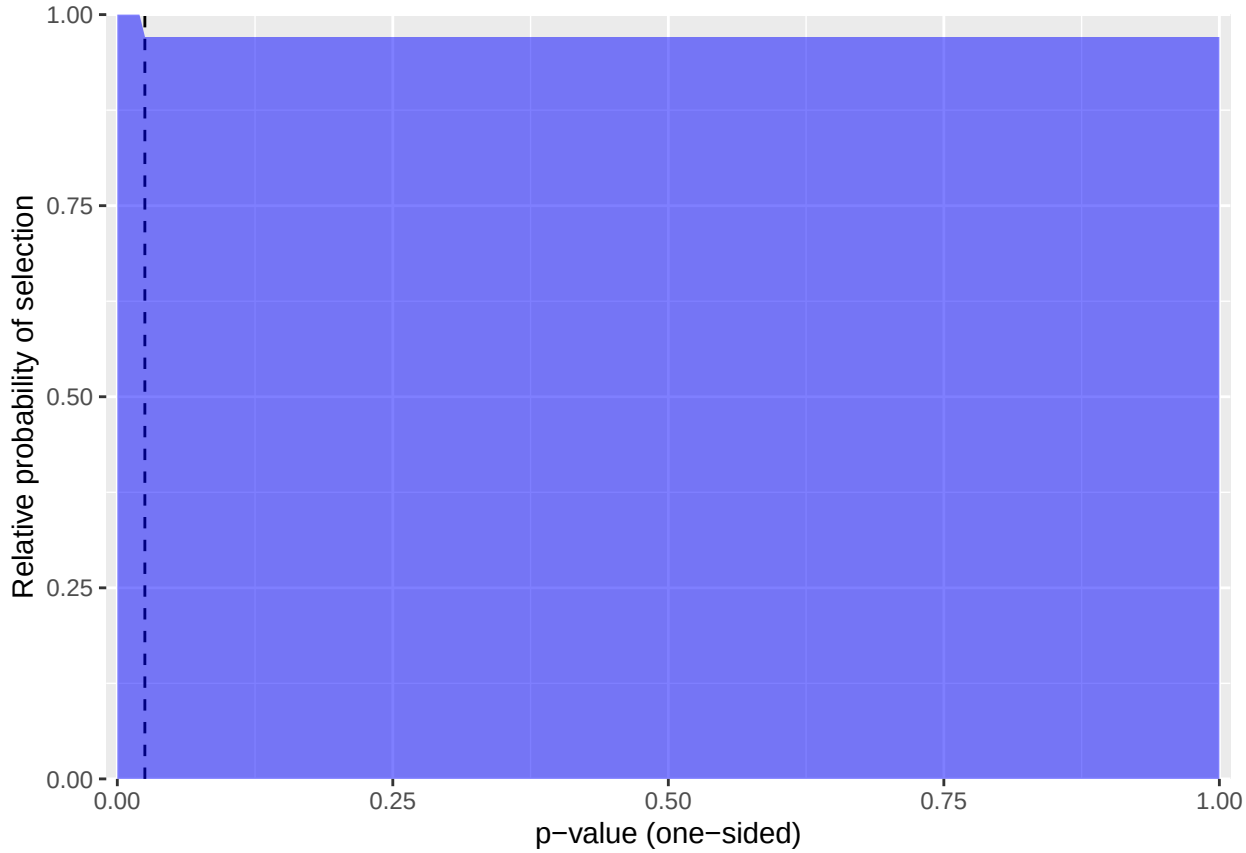
```
ex_adj_ms_step3_mod <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
```

```r
  steps = 0.025,
  mean_mods = ~ design
)

summary(ex_adj_ms_step3_mod)
```

```
## Step Function Model
##
## Call:
## selection_model(data = dat_ex, yi = d, sei = sd_d, cluster = studyid,
##     selection_type = "step", steps = 0.025, mean_mods = ~design)
##
## Number of clusters = 128; Number of effects = 251
##
## Steps: 0.025
## Estimator: composite marginal likelihood
## Variance estimator: robust
##
## Log composite likelihood of selection model: -41.84831
## Inverse selection weighted partial log likelihood: 189.7294
##
## Mean effect estimates:
##                                                  Large Sample
##             Coef. Estimate Std. Error  p-value Lower  Upper
##  beta_(Intercept)    0.251     0.0451 2.59e-08 0.163  0.339
##       beta_design    0.233     0.0494 2.44e-06 0.136  0.330
##
## Heterogeneity estimates:
##                                     Large Sample
##  Coef. Estimate Std. Error p-value  Lower  Upper
##   tau2   0.0746    0.00875     --- 0.0593 0.0939
##
## Selection process estimates:
##  Step: 0 < p <= 0.025; Studies: 106; Effects: 198
##                                      Large Sample
##    Coef. Estimate Std. Error p-value Lower  Upper
##  lambda0        1        ---     ---   ---    ---
##
##  Step: 0.025 < p <= 1; Studies: 38; Effects: 53
##                                      Large Sample
##    Coef. Estimate Std. Error p-value Lower  Upper
##  lambda1    0.971      0.291   0.921 0.539   1.75
```

### Selection plot: Step 3 PSM model moderator

```r
selection_plot(ex_adj_ms_step3_mod)
```




### Adjusted estimate in metaselection: Step 4PSM model moderator

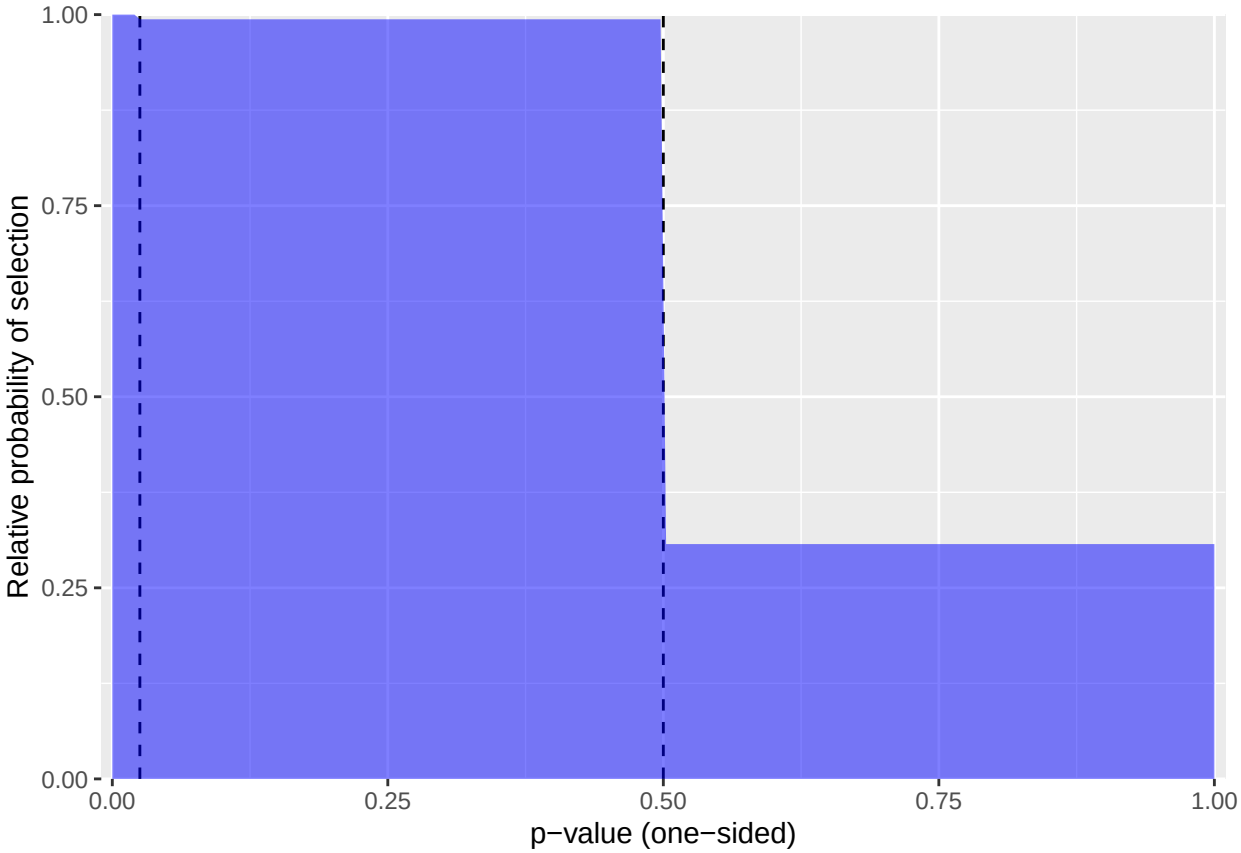
```
ex_adj_ms_step4_mod <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = c(0.025, .5),
  mean_mods = ~ design
)

summary(ex_adj_ms_step4_mod)
```

```
## Step Function Model
##
## Call:
## selection_model(data = dat_ex, yi = d, sei = sd_d, cluster = studyid,
##     selection_type = "step", steps = c(0.025, 0.5), mean_mods = ~design)
##
## Number of clusters = 128; Number of effects = 251
##
## Steps: 0.025, 0.5
## Estimator: composite marginal likelihood
## Variance estimator: robust
##
```

```
## Log composite likelihood of selection model: -35.85139
## Inverse selection weighted partial log likelihood: 193.7922
##
## Mean effect estimates:
##                                                  Large Sample
##             Coef. Estimate Std. Error  p-value  Lower  Upper
##  beta_(Intercept)    0.154     0.0646 1.68e-02 0.0279  0.281
##       beta_design    0.299     0.0660 6.12e-06 0.1692  0.428
##
## Heterogeneity estimates:
##                                     Large Sample
##  Coef. Estimate Std. Error p-value  Lower  Upper
##   tau2   0.0925     0.0123     --- 0.0713   0.12
##
## Selection process estimates:
##  Step: 0 < p <= 0.025; Studies: 106; Effects: 198
##                                      Large Sample
##    Coef. Estimate Std. Error p-value Lower  Upper
##  lambda0        1        ---     ---   ---    ---
##
##  Step: 0.025 < p <= 0.5; Studies: 32; Effects: 40
##                                      Large Sample
##    Coef. Estimate Std. Error p-value Lower  Upper
##  lambda1    0.994      0.246    0.98 0.612   1.61
##
##  Step: 0.5 < p <= 1; Studies: 9; Effects: 13
##                                      Large Sample
##    Coef. Estimate Std. Error p-value Lower  Upper
##  lambda2    0.307      0.162  0.0248  0.11  0.861
```

### Selection plot: Step 4PSM model moderator

```r
selection_plot(ex_adj_ms_step4_mod)
```


1.00
0.75
0.50
0.25
0.00
Relative probability of selection
0.00
0.25
0.50
0.75
1.00
p−value (one−sided)